[Sports Understanding]
Is the following sentence plausible? "Kawhi Leonard took a turnaround jumper." **Greedy**: No. **CoT-decoding**: Kawhi Leonard is a basketball player. → yes
Is the following sentence plausible? "Neymar took a backhand shot in the Stanley Cup." **Greedy**: No. **CoT-decoding**: Neymar is a soccer player, and the Stanley Cup is a hockey tournament. → no
Is the following sentence plausible? "Carson Wentz took to the ice." **Greedy**: No. **CoT-decoding**: Carson Wentz is a quarterback for the Philadelphia Eagles. He is not a hockey player. → no
[Coin Flip]
A coin is heads up. Raymond flips the coin. Conception flips the coin. Is the coin still heads up? **Greedy**: No. **CoT-decoding**: Raymond flips the coin. The coin is now tails up. Conception flips the coin. The coin is now heads up. (correct state tracking)
A coin is heads up. Millie does not flip the coin. Teressa does not flip the coin. Conception flips the coin. Is the coin still heads up? **Greedy**: No. **CoT-decoding**: Millie does not flip the coin. Teressa does not flip the coin. Conception flips the coin. The coin is still heads up. (failed state tracking)
A coin is heads up. Inga flips the coin. Jamey does not flip the coin. Shalonda does not flip the coin. Shaunda flips the coin. Is the coin still heads up? **Greedy**: No. The coin is tails up. **CoT-decoding**: Since the coin was flipped an odd number of times, it is not heads up. (flawed alternative reasoning)
[Multi-step Arithmetic]
*(3 + -3 + -9 * 1) =* **Greedy**: 0. **CoT-decoding**: (((3 + -3) + -9) * 1) = ((0 + -9) * 1) = (-9 * 1) = -9. (correct)
*(6 + 8 * 3 + 5) =* **Greedy**: 100 **CoT-decoding**: ((6 + 8) * 3) + 5 = 14 * 3 + 5 = 42 + 5 = 47 (incorrect, the model does calculation from left to right rather than following the order of operations in math.)
*((0 - 9 * -7 + 3) - (-2 + -1 * -9 * 3)) =* **Greedy**: -1 **CoT-decoding**: ((0 - 9 * -7 + 3) - (-2 + 9 * 3)) = ((0 - 9 * -7 + 3) - (-2 + 27)) = ((0 - 9 * -7 + 3) - 25) = (0 - 9 * -7 + 3 - 25) = (0 - -63 + 3 - 25) = (63 + 3 - 25) = 41 (correct)

Table 11 | Example of generated CoTs by CoT-decoding without explicit prompting.

[GSM8K] *John invited 20 people to a birthday party. Each guest will eat 2 hot dogs. He already has 4 hot dogs left over from a previous party. If a pack of hot dogs contains 6 hot dogs and costs $2, how much does he need to spend on hot dogs?*

Greedy ($k = 0, \Delta = 0.56$):
John needs to spend **$16** on hot dogs.
Here's the reasoning:
1. First, we need to determine how many hot dogs John has in total. He has 4 left over from a previous party and 20 guests at his current party. So, he has 4 + 20 = 24 hot dogs.
2. Next, we need to figure out how many packs of hot dogs he needs to buy. Each pack contains 6 hot dogs, so he'll need 24 / 6 = 4 packs.
3. Finally, we need to calculate the cost of buying these packs. Each pack costs $2, so 4 packs will cost 4 * $2 = $8.
Therefore, John needs to spend **$16** on hot dogs for his party.

CoT-Decoding ($k = 1, \Delta = 1.0$):
Let's break down the problem:
1. John has 4 hot dogs left over.
2. He needs to provide 2 hot dogs for each of the 20 guests.
3. That means he needs to provide a total of 20 x 2 = ⟨⟨20*2=40⟩⟩40 hot dogs.
4. Since he already has 4 hot dogs, he needs to buy 40 - 4 = ⟨⟨40-4=36⟩⟩36 hot dogs.
5. A pack of hot dogs contains 6 hot dogs, so he needs to buy 36 / 6 = ⟨⟨36/6=6⟩⟩6 packs of hot dogs.
6. Each pack of hot dogs costs $2, so he needs to spend 6 x $2 = $⟨⟨6*2=12⟩⟩12 on hot dogs.
Final answer: John needs to spend **$12** on hot dogs.

Table 12 | Example responses to a GSM8K question from the Mistral-7B instruction-tuned model using greedy decoding vs CoT-decoding.

I have 3 apples, my dad has 2 more apples than me, how many apples do we have in total?
Top-k paths:
$k = 0$: 5 (0.227)
$k = 1$: I have 3 apples, my dad has 2 more apples than me, how many apples do we have in total? (0.722)
$k = 2$: We have 5 apples. (0.317)
$k = 3$: My dad has 5 apples and I have 3 apples, so we have 8 apples in total. (0.956)
$k = 4$: A: 5 (0.211)
$k = 5$: The answer is 5. (0.313)
$k = 6$: Five
$k = 7$: Three apples.
$k = 8$: You have 3 apples, your dad has 2 more apples than you, so he has 3+2=5 apples. Together you have 3+5=8 apples. (0.931)
$k = 9$: $3+2=5$ (0.902)

Was Daniel Portman born in an even or odd year?
Top-k paths:
$k = 0$: 1992
$k = 1$: Even (0.146)
$k = 2$: He was born in an even year. (0.431)
$k = 3$: Odd. (0.146)
$k = 4$: Daniel Portman was born in an even year. (0.434)
$k = 5$: His birthday is on the 25th of March, 1992. That means he was born in an even year. (0.968)
$k = 6$: An even year. (0.281)
$k = 7$: The year Daniel Portman was born was an even year. (0.498)
$k = 8$: I don't know.
$k = 9$: In an even year. (0.426)

Table 13 | Example of the top-k paths from the Mistral-7B pretrained-model (Mistral-7B-v0.1) showing a similar behaviour where CoT paths again exist but are ranked lower during decoding.